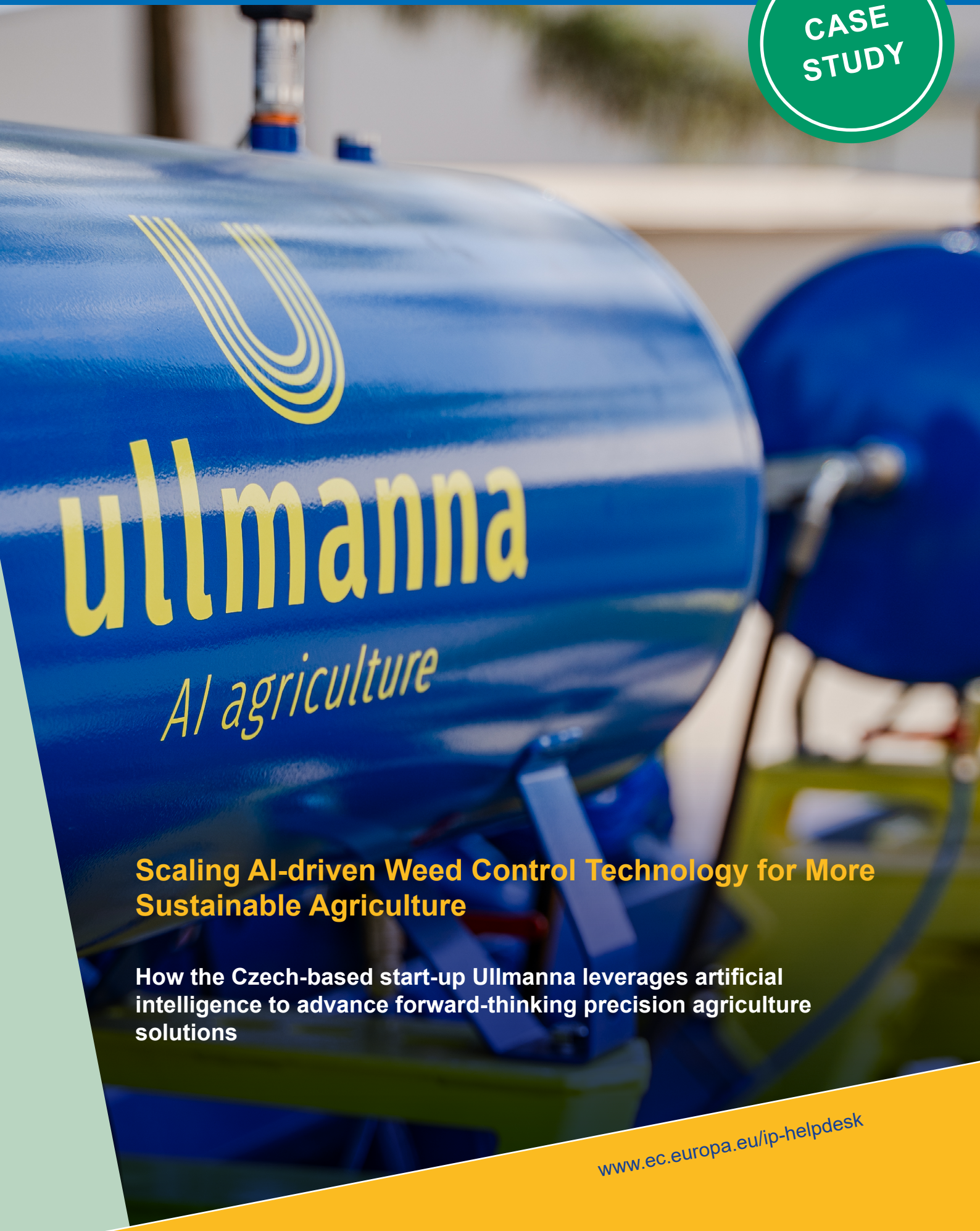
European
IP Helpdesk

CASE STUDY



Scaling AI-driven Weed Control Technology for More Sustainable Agriculture

How the Czech-based start-up Ullmanna leverages artificial intelligence to advance forward-thinking precision agriculture solutions

ullmanna
AI agriculture

Short company profile

Ullmanna s.r.o., a Czech-based start-up founded in 2019, leverages artificial intelligence (AI) to develop precision agriculture solutions, tackling key challenges faced by farmers. The company's goal is to revolutionise non-chemical weed control by providing farmers with scalable, cost-effective alternatives that reduce chemical use while maintaining productivity. Today, Ullmanna has a growing presence across Europe and Japan, with plans for expansion into the U.S., Middle East and Australian markets. The company is developing cutting-edge technology while also navigating the complex landscape of intellectual property (IP) protection to secure its innovation and ensure long-term success.



It's a family affair: the beginning of Ullmanna's journey

Agriculture is a highly demanding field where producers and other players in that field are faced with a number of regulations, specifically those related to the environment. On the other hand, countries are often not producing enough to satisfy their own needs. Hence, there is a constant battle between achieving high production rates and caring for the environment. Farmers often have to choose between chemical herbicides, which pose environmental and health risks, and manual weeding, which is costly and labour-intensive. Neither option is sustainable in the long run, especially as labour shortages became more intensified in the farming sector. This challenge increases the need to develop novel technologies and innovative solutions.

Against this backdrop, Ullmanna was founded in 2019 by Martin Ullmann, an IT expert, and his father-in-law, Jindřich Ullmann, an experienced agronomist with in-depth knowledge of farming challenges, especially in their local environment. Their collaboration began when Jindřich introduced Martin to the weed management issues faced by local farmers.



Jindřich Ullmann,

Chief Agriculture Officer & Co-founder of Ullmanna:

"As an agronomist, I saw firsthand the struggles farmers faced with weed management. Partnering with Martin, we combined agriculture and technology to create a sustainable, AI-driven solution. Seeing Ullmanna come to life has been incredibly rewarding."



Martin Ullmann,

Chief Executive Officer & Co-founder of Ullmanna:

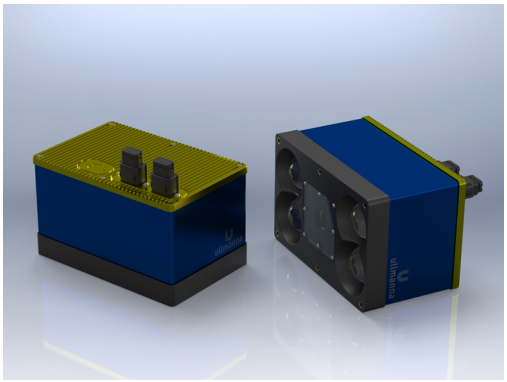
"Farmers were facing a challenge without a viable solution, and I fell in love with the idea that technology could revolutionise agriculture."

Seeing the challenges up close, Martin became deeply invested in finding a technological solution. He recognised the potential for AI and automation to transform agriculture, bridging the gap between sustainability and efficiency. Inspired by this vision, the IT specialist left his career in banking software to fully dedicate himself to developing an AI-driven prototype, marking the beginning of Ullmanna's journey.

Ullmanna developed two core innovations: the **AROW box**, an AI-powered camera system for precise crop identification and weed management, and the **NEWMAN machine**, an automated weeding system optimised for large-scale farming. Unlike traditional methods that focus on cutting weeds, Ullmanna's approach emphasises preserving crops and managing soil conditions to suppress weed regrowth, a method proven to be more effective and sustainable in the long term.

However, along its journey, the young family business faced several challenges linked to technological development and validation, market adoption and IP protection.

THE ARROW BOX



THE CULTIVATOR

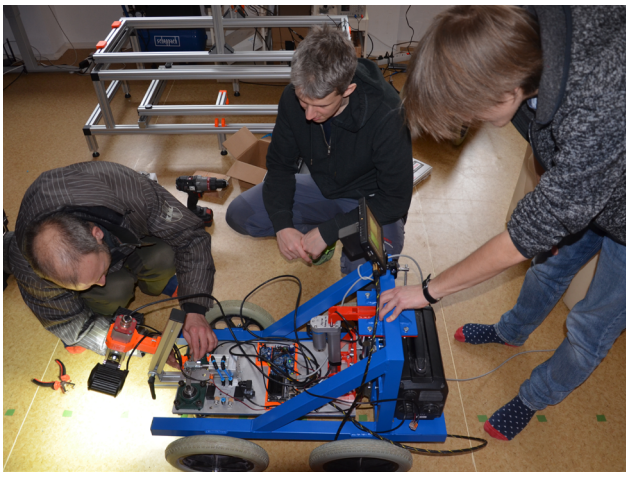


From prototype to market readiness: significant steps in Ullmanna's company development

Ullmanna's journey began with the development of a Minimum Viable Product (MVP), which was a simple demonstrator of a weeder that recognised only green dots, without any AI capabilities. While this prototype validated the basic concept, it was clear that a more sophisticated approach was needed for precise crop recognition in real-world farming conditions.

Recognising this, Ullmanna collaborated with university researchers to explore the viability of integrating AI for advanced crop and weed detection. This partnership provided the foundation for further developing their AI-driven technology. They first approached the Centre for Innovation Support at the Technical University of Ostrava, where they received early guidance on IP strategy, business planning, and funding opportunities. This was a crucial first step, because it encouraged them to think on a larger scale, refining their vision and business model. The innovation centre also helped them prepare a proposal for a small local grant, which provided the financial means to establish Ullmanna as a formal company.

Turning the concept into a market-ready solution required significant research and development (R&D) efforts. The challenge was to develop a precise, scalable system capable of accurately identifying crops and weeds while operating efficiently under real-life farming conditions. This demanded ongoing refinement, validation and testing to ensure precision, adaptability and reliability across different soil types, crops, and weather conditions.



However, one of the challenges in technology validation was seasonality. Farmers have a limited window each year (typically spring) to adopt and test new technologies. Any delays in product readiness could postpone the implementation for an entire year, hampering market adoption and entry. Additionally, many farmers were accustomed to traditional methods and often hesitant to adopt new, not yet established technologies. High initial investment costs posed another obstacle, as many required financial incentives, subsidies, or clear economic benefits before committing to new, high-tech farming equipment.

Ullmanna's first major breakthrough came with the EU-funded project "agROBOfood" (2019-2020) running under the EU's Funding Programme for Research & Innovation "Horizon 2020", aimed at establishing sustainable network of Digital Innovation Hubs (DIHs) and connecting them with small and medium enterprises (SMEs). At the heart of this project, certain innovation experiments were organised and monitored by those DIHs, which secured funds of €460,000 and allowed them to expand their team, and conduct large-scale testing across multiple European farms. The funding played a key role in developing and refining the company's technology: the NEWMAN machine and the AROW box.

During the early phase of R&D, Ullmanna realised that the AROW box had significant potential beyond its integration into the NEWMAN machine. While tested as part of the complete system, the AROW box's AI-powered vision technology could be integrated into various weeding machines produced by other manufacturers, increasing the company's scalability and market reach. This discovery led Ullmanna to shift its focus from manufacturing entire machines to specialising in AI-powered camera technology, making the AROW box its core differentiator in the market.

This shift was further supported by additional funding obtained from the European Institute of Innovation and Technology (EIT) Food which played a critical role for validation across multiple phases: testing with farmers abroad through the EIT Test Farm (2021); participation in the EIT Food Accelerator Network (2022); and scale-up funding through the EIT Food Open Innovation call (2023-2024). In the latter phase, Ullmanna participated in a consortium that included two farms, two manufacturers, and a research institution.

These activities provided instrumental support in transforming the company from large-scale validation to commercial adoption. By the end of this phase, the company had signed commercial agreements with several manufacturers, marking an important step toward market deployment.

HOW IT WORKS?



1. Camera detects the drop and the flashlights provide consistent light for best recognition
2. AI sees multiple pictures of one plant and calculates about 5-6 times if it is a plant or not
3. If it is a plant: Knives open
4. After the plant: Knives close
5. Farmer is happy: Weed is gone, plant has more space for growing

Securing IP and gaining competitive advantage

Apart from addressing technology development and market adoption hurdles, safeguarding Ullmanna's innovation from the outset was equally critical. As one of the first companies to integrate AI-driven crop and weed recognition into precision agriculture, Ullmanna faced the risk of competitors replicating its technology. However, when the company first consulted a patent attorney, they were advised that their software-based AI approach might not be patentable. The attorney also pointed out that parts of their innovation had already been published online. With novelty being a fundamental condition to securing patents, this created significant uncertainty for Ullmanna, as without a strong patent, their technology remained vulnerable to imitation, particularly as the company was preparing for international expansion.

Ullmanna's shift toward AI-powered camera technology made securing strong IP protection increasingly important. This led to a critical turning point in the company's strategy. Determined to protect its technology, Ullmanna sought additional support, leading to a critical breakthrough in its IP strategy. Through local and national mentorship programmes focused on IP protection, the company was introduced to another patent attorney with expertise in emerging technologies. Unlike the initial assessment, the second attorney conducted a deeper technical evaluation, identifying a key patentable component: the steering knives and mechanical integration of AI-powered vision technology.



This distinction set Ullmanna's system apart from existing solutions and laid the foundation for a strong intellectual assets strategy based on patent protection in most important markets. Thanks to the funding support by EIT Food, Ullmanna proceeded with a national patent application in the Czechia, to secure priority date for the invention. In the next phase, after a period of 12 months, company submitted a Patent Cooperation Treaty (PCT) application to extend protection to key international markets, including European Patent Office (EPO) member countries, the United States, Canada, Australia, and South Africa. The patent is currently granted in Czechia and South Africa, a strategically

important market where many farmers source machinery from Europe. EPO has recently completed the examination process, and the patent is expected to be granted shortly also in EPO member states. Granting patents in the remaining countries are still in process. This multi-country approach reflects Ullmanna's commitment to securing its innovations as it scales globally.

As Ullmanna grew and entered new markets, navigating international IP regulations became an even greater challenge. Each country had different enforcement mechanisms, making global IP protection complex, time-consuming, and costly. For a start-up with limited resources, the high costs of securing patents had to be weighed against the urgency of bringing products to market and generating revenue. These challenges required strategic decisions about where to prioritise IP filings, ensuring that Ullmanna's most critical innovations were safeguarded while continuing to scale operations. Nevertheless, the granted patents provided a critical competitive edge, allowing the company to protect its core technology while exploring additional funding opportunities for further patent filings.

In addition to securing patent protection, Ullmanna implemented a trade secret strategy to safeguard aspects of its technology that were not easily patentable. This includes its data sets, source code, and the set-up of the electronic control boards. This approach ensured that the underlying AI models and machine learning algorithms used to improve weed detection remained proprietary, preventing easy replication by competitors. By combining formal IP rights with confidential technological components, Ullmanna created a multi-layered protection strategy, securing its innovations while retaining the flexibility to adapt to new markets, collaborate with manufacturers, and continuously enhance its AI-driven system.

The company's technology is now actively used in Germany, Austria, Switzerland, France, Italy and Czechia, with additional deployments in Japan, where precision farming solutions are in high demand. As part of its long-term growth strategy, Ullmanna is evaluating opportunities for expansion into the U.S. and Australia, ensuring that IP protections and strategic partnerships are in place before entering these new markets.

Lessons Learned

Ullmanna's journey from concept to market-ready innovation offers valuable insights for start-ups navigating technology development, IP protection, commercialisation, and sustainability-driven markets.

- Securing **public funding was essential for R&D** and market validation. As a start-up, Ullmanna lacked the resources for large-scale testing, but grants from Horizon 2020 agROBOfood and EIT Food enabled critical advancements in AI, automation, and commercialisation while reducing financial risks.

- Crafting a **balanced IP strategy combining patents and trade secrets** provided cost-effective and more comprehensive protection. Despite initial challenges, Ullmanna successfully secured an international patent with 23 approved claims while also safeguarding proprietary AI models and hardware components through trade secrets. This dual approach ensured protection without overextending resources on international filings.

- **Aligning with sustainability trends** created funding and market opportunities. As regulations tightened on chemical herbicides, Ullmanna's non-chemical weed control solution positioned the company at the forefront of an emerging market, securing investor interest and regulatory support.

- **Collaborating with key stakeholders** during validation helped overcome adoption barriers. Farmers, often hesitant due to financial constraints, benefited from subsidised access through EIT Food projects, facilitating trust and market entry. Close partnerships with manufacturers and early adopters also ensured real-world refinement and credibility.

- Finally, **interdisciplinary expertise** was a major driver of success. Ullmanna's fusion of IT (AI, automation) and agricultural knowledge resulted in a highly practical yet technologically advanced solution. Ongoing collaborations with manufacturers and farmers continue to bolster its innovation processes and expansion.

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